Exercise 12.3 SOLUTION

Q1. A substance decays exponentially following the formula $A(t) = A_0 e^{-0.1t}$, where A_0 is the initial amount. Find the limit of $A(t)$ as $t \to \infty$.

Sol:

$$A(t) = A_0 e^{-0.1t} = A_0 \lim_{t\to\infty}\left(\frac{1}{e^{0.1t}}\right)$$

$$= A_0 . \frac{1}{e^{\infty}}$$

$$= A_0 \frac{1}{\infty}$$

$$= A_0 (0) = 0$$

Q2. A town's population is modeled by $P(t) = \frac{100000}{1+9e^{-0.5t}}$. what is the long-term population as $t \to \infty$.

Sol:

$$P(t) = \frac{100000}{1+9e^{-0.5t}}$$

$$\lim_{t\to\infty} P(t) = \lim_{t\to\infty} \frac{100000}{1+9e^{-0\,5t}}$$

$$= \frac{100000}{1+\frac{9}{e^{\infty}}} = \frac{100000}{1+0} = 100000$$

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Q3. A company's weekly sales (in thousands) follow the functions $S(t)=\dfrac{500t}{t+10}$. what is the limit of $S(t)$ as $t\to\infty$ and what does it represent?

Sol:

$$S(t)=\frac{500t}{t+10}$$

$$\lim_{t\to\infty}S(t)=\lim_{t\to\infty}\frac{500t}{t+10}$$

$$=\lim_{t\to\infty}\frac{500\not{t}}{\not{t}\left(1+\frac{10}{t}\right)}=\left(\frac{500}{1+\frac{10}{\infty}}\right)=\frac{500}{1+0}=500$$

Q4. Signal strength $S(d)$ at a distance d from a tower modeled as $S(d)=\dfrac{1000}{d^2}$.

i. what is the signal at $d=10$?

Sol:

$$S(10)=\frac{1000}{(10)^2}=\frac{1000}{100}=10$$

ii. What happens to signal strength as $d\to\infty$?

Sol:

$$\lim_{d\to\infty}\frac{1000}{d^2}=\frac{1000}{\infty^2}=\frac{1000}{\infty}=0$$

Q5. A stock price grows according to the function $P(t)=50e^{0.05t}$

i. Find the limit of $P(t)$ as $t\to\infty$

Sol:

$$\lim_{t\to\infty}P(t)=\lim_{t\to\infty}50e^{0.05t}=50e^{\infty}=(50)\infty$$

ii. Calculate the price of ten 10 years.
Put t = 10

Sol:

$$P(10)=50e^{0.05(10)}=82.44$$

Q6. The factory's cost function is given as:

$$C(x)=\begin{cases}10x+500 & if \quad x\le 100\\ 12x+300 & if \quad x>100\end{cases}$$

In the cost function continuous at $x=100$?

Sol:

$$C(x)=\begin{cases}10x+500 & if \quad x\le 100\\ 12x+300 & if \quad x>100\end{cases}$$

$$C(100)=10(100)+500=(1000+500)=1500$$

$$\text{L.H.L}=\lim_{x\to 100^-}f(x)$$

$$=\lim_{x\to 100^-}(10x+500)$$

$$=10(100)+500=1000+500=1500$$

$$\text{R.H.L}=\lim_{x\to 100^+}C(x)=\lim_{x\to 100^+}(12x+300)$$

$$=12(100)+300=1200+300=1500$$

$$\Rightarrow\lim_{x\to 100}C(x)=C(100)$$

$$\Rightarrow C(x)\text{ is continuous at }x=100$$

Q7. Inflation is modeled by $I(t)=I_0e^{0.03t}$, where I_0 is the initial price index and t is the number of years

i. Find the inflation rate after 5 years if $I_0=100$

Sol: Inflation rate after 5 years if $I_0=100$

$$I_0=100\quad,\quad t=5$$

$$I(5)=100e^{0.03(5)}=(100)(1.11618)=116.18$$

$$\text{Inflation Rate}=\left(\frac{I-I_0}{I_0}\right)\times 100\%$$

$$=\left(\frac{116.18-100}{100}\right)\times 100\%=16.18\%$$

ii. What is the expected price index after 10 years

Sol: Expected Price index after 10 years

$$I(10)=100e^{0.03(10)}=134.99$$

Q8. The cost to produce x units is:

$$C(x)=\begin{cases}5x+20 & if \quad x\le 10\\ 6x+10 & if \quad x>10\end{cases}$$

If the cost function at $x=10$?

Sol:

$$C(x)=\begin{cases}5x+20 & if \quad x\le 10\\ 6x+10 & if \quad x>10\end{cases}$$

$$C(10)=5(10)+20=50+20=70$$

$$\text{L.H.L}=\lim_{x\to 10^-}C(x)=\lim_{x\to 10^-}(5x+20)$$

$$=5(10)+20=50+20=70$$

$$\text{R.H.L}=\lim_{x\to 10^+}C(x)=\lim_{x\to 10^+}(6x+10)$$

$$=6(10)+10=60+10=70$$

$$\Rightarrow\lim_{x\to 10}C(x)=C(10)$$

$$\Rightarrow\lim_{x\to 10}C(x)\text{ is continuous at }x=10$$

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